

Introduction

DNA Sequencing process

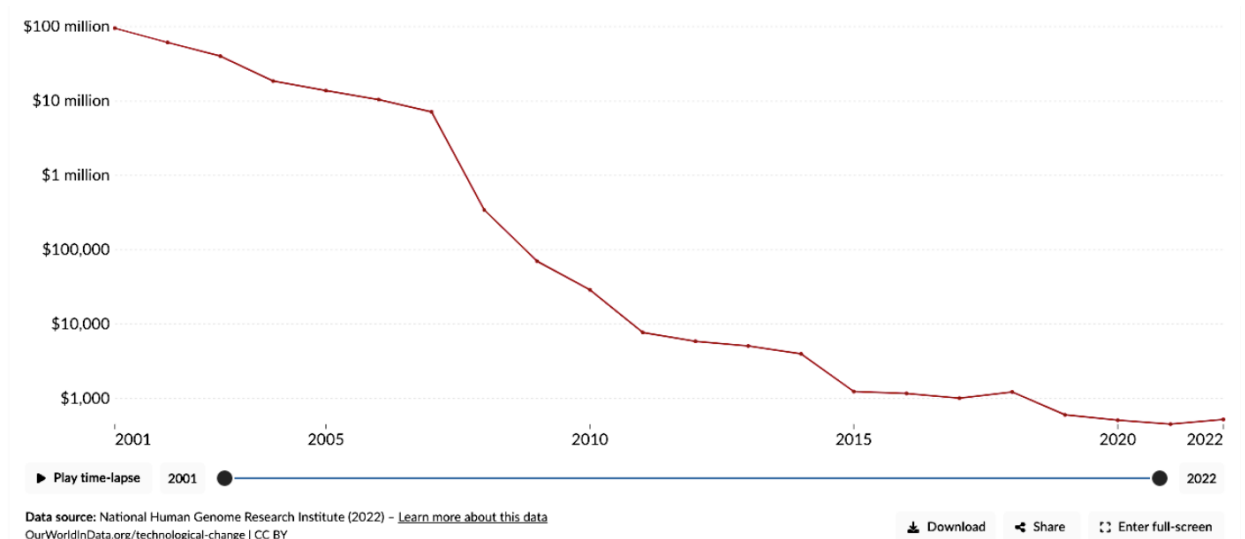
- Process of determining the order of nucleotides in a genome
- Sanger Sequencing - first-generation sequencing methods
 - Produces DNA fragments of varying lengths terminated at specific bases
 - Fragments separated by size and read via fluorescent tags

Next Generation Sequencing (NGS)

- Increasing the throughput by massively parallelising the sequencing process

NGS advantages over Sanger

- High throughput: Faster turnaround times
- Cost-effective: Reduced human genome sequencing from millions of dollars to affordable levels



<https://ourworldindata.org/grapher/cost-of-sequencing-a-full-human-genome>

- Large-scale applications: Enables population-scale studies and generates terabytes of data
- Modern accuracy: >99% variant detection accuracy (improved from early NGS limitations)
- Versatile read lengths: Both short-read and long-read platforms available

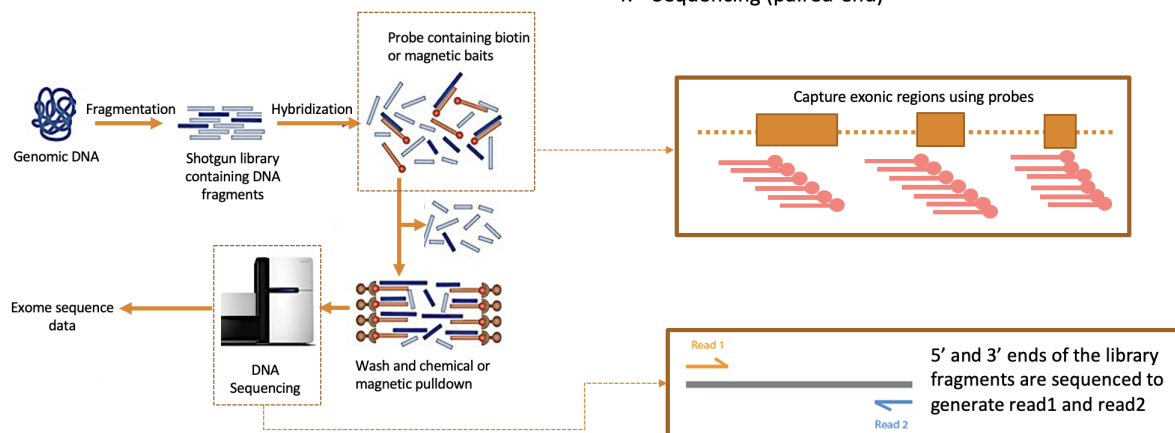
NGS - Exome vs Whole-Genome Sequencing

Exome Sequencing (WES)

- Focuses on protein-coding regions (~2% of genome)
- Contains ~85% of disease-causing mutations in Mendelian disorders
- Cost-effective tool for genetic disease analysis

Exome sequencing workflow

1. Fragmentation and library preparation
2. Hybridization to exonic regions (regions in exome capture array)
3. Magnetic pull down and wash excess fragments
4. Sequencing (paired-end)



Output - Raw sequence data (FASTQ)

FASTQ files from Illumina sequencing technology list "reads" from the sequencing machine - Read-1 and Read-2 FASTQ files. Each entry (i.e., single read) in a FASTQ file contains 4 lines

- First line: A sequence identifier with metadata on the sequencing run the read
- The sequence (i.e., the base calls)
- A separator
- Quality scores of each base call.

Single entry in a FASTQ file

```
@ERR059938.60 HS9_6783:8:2304:19291:186369#7/1
GTCTCCGGGGGCTGGGGGAACCAAGGGGTTCCCAACCAACCACCTCACTCAGCCTTTTCC
CTCCAGGCATCTCTGGGAAAGGACATGGGGCTGGTGCGGGG
+
7?CIGJB:D:-F7LA:GI9FDHBIJ7,GHGJBKHNI7IN,EML8IFIA7HN7J6,L6686LCJE?
JKA6G7AK6GK5C6@6IK+++?5+=<;227*6054
```

Additional reading materials

- [FASTQ files explained from knowledge.illumina.com](https://www.illumina.com/sequencing/technology/next-generation/FASTQ_files_explained_from_knowledge.illumina.com)
- [About FASTQ sequence read files from IGSR resources](https://www.illumina.com/sequencing/technology/next-generation/About_FASTQ_sequence_read_files_from_IGSR_resources)

Key Challenges of WES

- Limited to protein-coding regions (disease-relevant non-coding variants)

Whole-Genome Sequencing

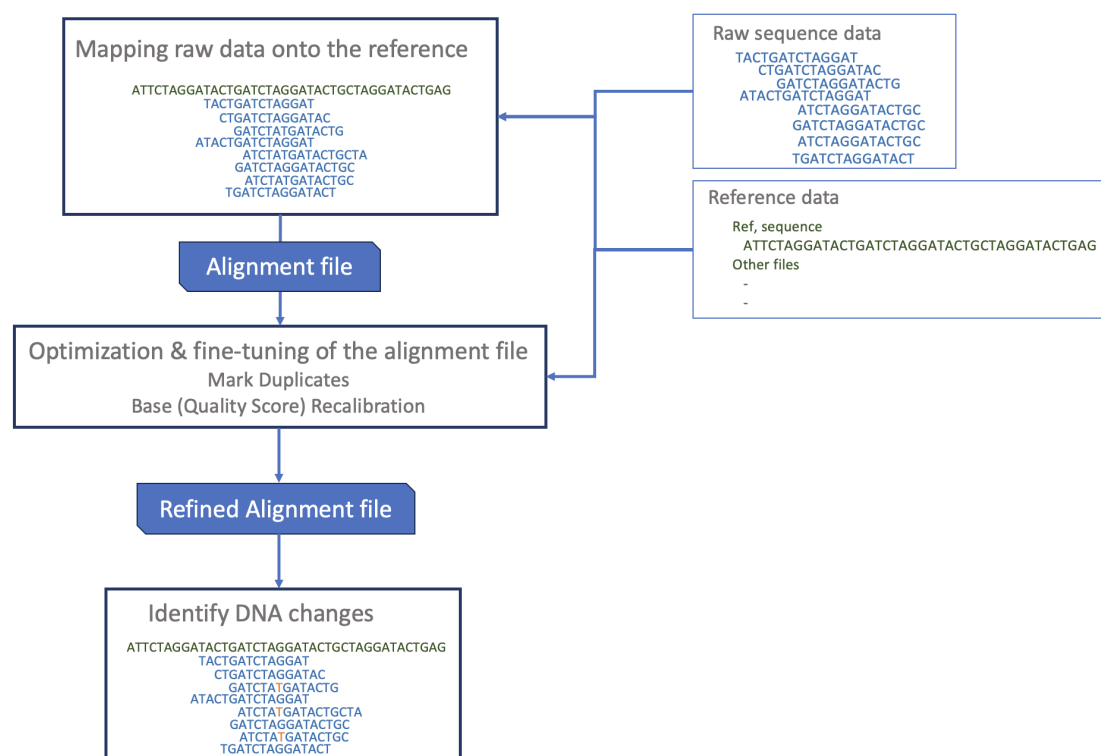
- Analyze the whole genome, including coding, non-coding, and mitochondrial DNA
- Discover a broad range of variants

Whole Genome Sequencing (WGS) Challenges

- Data Volume & Computational Burden
 - Massive data: Individual samples: dozens to 250 GB; daily output: terabytes
 - CPU limitations: Traditional processors can't keep pace with data generation
 - Inefficient parallelization: Scatter-and-gather overheads, only 2-3x speedup despite increased resources
- Speed & Turnaround Time
 - Slow processing: Standard WGS takes days (over 24 hours) per sample on CPU-only systems
 - Clinical urgency: this turnaround is insufficient for critically ill patients

NGS data processing workflow

A basic NGS processing workflow comprises several key steps.



- Mapping raw data onto the reference
- Optimization & fine-tuning of the alignment file
 - Mark Duplicates

- Base (Quality Score) Recalibration
- Identify DNA changes

Main steps in a germline variant calling (short variants - InDel and SNP calling)

Basic best-practices NGS processing workflow (CPU-native)

A GATK best-practices workflow runs several GATK tools to call SNVs and InDels from sequence datasets

- Read mapping and alignment refinement stages:
 - Step 1: BWA read mapping and sorting
 - Step 2: GATK MarkDuplicates
 - Step 3: GATK BaseRecalibrator
 - Step 4: GATK ApplyBQSR
- SNVs and InDels calling can be performed by using a caller like HaplotypeCaller (Step 5)